MATH317 · NUMERICAL METHODS

What repeats in the midterm

Analysis of four midterms: 461 · 462 · 471 · 472

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What's inside

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01 The one-minute summary

The numbers change every term, but the shape of the questions barely does.

Three questions cover most of the marks

Floating-point and **Bisection** appeared in all four. **Horner + Taylor** appeared in three.

The midterm is 60 minutes and 30 marks (461 was out of 25).

02 Frequency table

Topic	461	462	471	472	Count	Marks
Floating-point representation and hex	✓	✓	✓	✓	$\frac{4}{4}$	6–7
Root existence, then bisection with n_max = 2	✓	✓	✓	✓	$\frac{4}{4}$	8
Horner, then Taylor about r		✓	✓	✓	$\frac{3}{4}$	8
Absolute and relative error	✓	✓	✓		$\frac{3}{4}$	4–7
True / False				✓	$\frac{1}{4}$	7
Number of bisection iterations		✓			$\frac{1}{4}$	3
Newton, two iterations	✓				$\frac{1}{4}$	5

03 Details of each question

Floating-point

Midterm	Number	Type	Answer
472	`-23.21875`	Single	`C1B9C000`
471	`-45.21875`	Single	`C234E000`
462	`-184.125`	Double	`C067040000000000`
461	`1259.125`	Single	`449D6400`

The fractions are always `.125` or `.21875`, which end after one or two octal digits. Three of the four numbers are negative.

Horner + Taylor

Midterm	P(x)	Horner	Taylor
472	$3x^3 - 4x^2 + 22$	$x = -2$	$c = -3$
471	$3x^3 - 4x^2 - 12$	$x = -2$	$r = 3$
462	$4x^3 - 8x^2 + 14x - 32$	$x = 3$	$r = 2$

Always a cubic. In two of the three the x term is missing, so a zero must be written in its place.

Bisection

Midterm	f(x)	Interval
472	$-x^2 + 4x - e^x$	[0, 1]
471	$x^3 - 9x + 1$	[2, 4]
462	$2x^2 - e^x$	[0, 2]
461	$e^x - 4x^2$	[0, 1]

The stopping rule is identical in all four: `n_max = 2` or an error below 0.5×10^{-6}. So there are always three rows. Three of the four functions contain e^x.

04 What to study first

Priority	Topic	File
1	Floating-point conversion (single and double)	Chapter 2
2	Bisection with the table	Chapter 3.1
3	Horner and complete Horner	Chapter 1.1 · 1.2
4	Absolute and relative error	Chapter 1.1
5	Concepts (True/False)	All chapters
6	Newton	Chapter 3.2

Caution

This order is based on only four papers. A topic that did not appear before can still appear, especially Newton, since Lecture 6 is in your stated scope.

05 Mistakes that cost marks

Mistake	Fix
Skipping a step	The key gives marks per step. Write every line.
Forgetting the zero for a missing term in Horner	Write all coefficients first.
Storing the leading 1 in the mantissa	Only the bits after the point.
Using 127 with double precision	1023 and an 11-bit exponent.
Calculator in degree mode	Radian.